Lecture 11

# Poisson manifolds and the rigid body

**Objectives.** *Lecture 10 divided symmetries out of canonical systems. Dividing the rotations out of the rigid body leaves three variables, the components of its body angular momentum. Their bracket has a matrix $\Pi(x)$ that varies with the point and is singular everywhere. Its phase spaces are the spheres centred at the origin, together with the origin itself. The body angular momentum $\mathbf{M}$ evolves on a sphere. For three distinct principal moments of inertia, the rotations about the axes of least and greatest inertia are stable on each nonzero sphere, and the middle-axis rotations are unstable.*

By the end of this lecture you will be able to:

- Compute Lie–Poisson brackets and their Casimirs.
- Describe the canonical charts of a nonzero angular-momentum sphere.
- Classify steady rotations for three distinct principal inertias by stability on that sphere.

## 11.1 Poisson structures on $\mathbb{R}^m$

Since Lecture 5 a Hamiltonian system has lived on an open subset of $\mathbb{R}^{2n}$ with canonical coordinates, $\{q_i, p_j\} = \delta_{ij}$ and all other brackets zero. We now keep the algebra of Lecture 5 — bilinear, antisymmetric, Leibniz, Jacobi — and allow coordinates that are not canonical.

**Definition 11.1** (Poisson structure)**.** Let $U \subseteq \mathbb{R}^m$ be open. A *Poisson structure* on $U$ is a smooth map $\Pi\colon U \to \mathbb{R}^{m\times m}$ with $\Pi(x)^{\mathsf{T}} = -\Pi(x)$ for every $x$, such that the bracket

$$\{f, g\}(x) = \nabla f(x)^{\mathsf{T}}\,\Pi(x)\,\nabla g(x) = \sum_{a,b=1}^{m} \Pi_{ab}(x)\,\frac{\partial f}{\partial x_a}\frac{\partial g}{\partial x_b} \tag{11.1}$$

satisfies the Jacobi identity on $C^\infty(U)$. We call $\Pi$ the *structure matrix* and $(U, \Pi)$ a *Poisson manifold*. The *Hamiltonian field* of $f$ is $X_f = \Pi\nabla f$.

Three things are new. The dimension $m$ need not be even. The matrix $\Pi(x)$ need not be invertible. And $\Pi$ may vary from point to point. The old bracket is the case $m = 2n$, $\Pi = \mathbb{J}$.

The algebraic rules are unchanged. Putting $f = x_a$ and $g = x_b$ in (11.1) gives

$$\{x_a, x_b\} = \Pi_{ab}, \tag{11.2}$$

so the structure matrix *is* the table of brackets of the coordinates. Bilinearity and antisymmetry are immediate, and so is Leibniz ($\{f, gh\} = \{f, g\}\,h + g\,\{f, h\}$), because (11.1) differentiates $g$ once. Only the Jacobi identity is a condition, and by Remark 5.6 it holds

for all observables as soon as it holds for the coordinate functions. As in Definition 6.1, $X_f h = \nabla h^{\mathsf{T}} \Pi \nabla f = \{h, f\}$, so the motion generated by $H$ is

$$\dot{x} = \Pi(x)\,\nabla H(x), \qquad \dot{f} = \{f, H\}\,, \tag{11.3}$$

and $H$ is conserved along it, since $\{H, H\} = 0$ by antisymmetry.

An invertible $\Pi(x)$ forces $m$ even, since $\det \Pi = \det \Pi^{\mathsf{T}} = (-1)^m \det \Pi$. This lecture is about the other case, a *singular* $\Pi(x)$.

## 11.2 The Lie–Poisson bracket, and where it comes from

We begin with the angular-momentum bracket.

**Definition 11.2** (The Lie–Poisson bracket)**.** On $\mathbb{R}^3$, with the point written $\mathbf{L} = (L_1, L_2, L_3)$, the *Lie–Poisson bracket* is

$$\{f, g\}\,(\mathbf{L}) = \mathbf{L} \cdot \big(\nabla f \times \nabla g\big), \qquad \text{that is} \qquad \Pi_{ab}(\mathbf{L}) = \sum_c \varepsilon_{abc} L_c. \tag{11.4}$$

The structure matrix is *linear* in $\mathbf{L}$, and its coordinate brackets are

$$\{L_1, L_2\} = L_3, \qquad \{L_2, L_3\} = L_1, \qquad \{L_3, L_1\} = L_2, \tag{11.5}$$

which is (5.11): the brackets of the three components of angular momentum. From (11.4), $X_f = \Pi \nabla f$ works out to

$$X_f(\mathbf{L}) = \nabla f(\mathbf{L}) \times \mathbf{L}. \tag{11.6}$$

In particular $X_{L_3} = e_3 \times \mathbf{L}$: *the third component generates the rotation about the third axis*, counter-clockwise seen from above and at unit angular speed.

**Proposition 11.3** (The Lie–Poisson bracket is a Poisson structure)**.** (11.4) *satisfies the Jacobi identity.*

*Proof.* By Remark 5.6 it is enough to check it on the coordinates. For constant $u, v \in \mathbb{R}^3$, (11.4) gives $\{u \cdot \mathbf{L}, v \cdot \mathbf{L}\} = (u \times v) \cdot \mathbf{L}$, so the cyclic sum of $\{\{u \cdot \mathbf{L}, v \cdot \mathbf{L}\}, w \cdot \mathbf{L}\}$ is $\big[(u \times v) \times w + (v \times w) \times u + (w \times u) \times v\big] \cdot \mathbf{L} = 0$, by the Jacobi identity of the cross product. Take $u, v, w = e_a, e_b, e_c$. ∎

This bracket is what the canonical bracket leaves behind when only the angular momentum is kept. To say so we need maps that carry one bracket to another.

**Definition 11.4** (Poisson map)**.** Let $(U, \Pi)$ and $(V, \Pi')$ be Poisson manifolds. A smooth $F\colon U \to V$ is a *Poisson map* if $\{f \circ F, g \circ F\}_U = \{f, g\}_V \circ F$ for all $f, g \in C^\infty(V)$.

**Theorem 11.5** ($q \times p$ is a Poisson map)**.** $J\colon \mathbb{R}^6 \to \mathbb{R}^3$, $J(q,p) = q \times p$, *is a Poisson map from the canonical bracket to* (11.4).

*Proof.* By the chain rule, for $f \in C^\infty(\mathbb{R}^3)$, $\partial(f \circ J)/\partial q_i = \sum_a (\partial_a f \circ J)\, \partial J_a/\partial q_i$, and likewise in $p$. Substituting into (5.1) and collecting,

$$\{f \circ J, g \circ J\} = \sum_{a,b} (\partial_a f \circ J)(\partial_b g \circ J)\ \{J_a, J_b\}\,.$$

By (5.11), $\{J_a, J_b\} = \sum_c \varepsilon_{abc} J_c$, so the right-hand side is $\big[\mathbf{L} \cdot (\nabla f \times \nabla g)\big] \circ J$ — the function of $\mathbf{L}$ in the square bracket, evaluated at $\mathbf{L} = J(q,p)$ — which is $\{f, g\} \circ J$ for the bracket (11.4). ∎

## 11.3 Casimirs and leaves

**Definition 11.6** (Casimir). $C \in C^\infty(U)$ is a *Casimir* of a Poisson structure if $\{C, f\} = 0$ for every $f \in C^\infty(U)$, equivalently if $X_C = 0$.

A Casimir is conserved by *every* Hamiltonian, since $\dot C = \{C, H\} = 0$ whatever $H$ is. Its Hamiltonian vector field vanishes, so its flow is the identity. In contrast, a nonconstant symmetry charge may generate a nontrivial flow. On a connected canonical phase space all Casimirs are constant: Proposition 5.8 showed that on a connected canonical phase space $\{C, f\} = 0$ for all $f$ forces $C$ constant, and the same argument works wherever $\Pi$ is invertible, since $X_C = \Pi\nabla C = 0$ then gives $\nabla C = 0$. Degeneracy is what makes the notion non-trivial.

**Proposition 11.7** (The Casimirs of (11.4)). *$C \in C^\infty(\mathbb{R}^3)$ is a Casimir of the Lie–Poisson bracket if and only if $\mathbf{L} \times \nabla C(\mathbf{L}) = 0$ for every $\mathbf{L}$, and that holds if and only if $C$ is constant on every sphere $|\mathbf{L}| = R$. In particular $|\mathbf{L}|^2$ is a Casimir.*

*Proof.* By the triple-product identity, $\{C, f\} = \mathbf{L} \cdot (\nabla C \times \nabla f) = \nabla f \cdot (\mathbf{L} \times \nabla C)$. This vanishes for every $f$ if and only if $\mathbf{L} \times \nabla C = 0$ (take $f = L_a$ for the three values of $a$). Suppose that holds. At $\mathbf{L} \neq 0$ it says $\nabla C(\mathbf{L})$ is a multiple of $\mathbf{L}$, so the derivative of $C$ in every direction tangent to the sphere through $\mathbf{L}$ is zero; joining two points of that sphere by a smooth path on it and differentiating $C$ along the path shows $C$ is constant on it. The sphere of radius 0 is a point. Conversely, if $C$ is constant on every sphere then its derivative along every tangent direction vanishes, so $\nabla C$ is radial and $\mathbf{L} \times \nabla C = 0$. Finally $\nabla |\mathbf{L}|^2 = 2\mathbf{L}$. ∎

So every motion, whatever its Hamiltonian, stays on the sphere $|\mathbf{L}| = R$ on which it starts. That sphere ought to be a phase space with one degree of freedom. But it is not an open subset of $\mathbb{R}^2$, and no single system of coordinates covers it.

### Phase space by canonical charts

By a *chart* on a set $S \subseteq \mathbb{R}^N$ we mean a bijection $\kappa : U \to V$ from a subset $U \subseteq S$ that is *open in $S$* (of the form $S \cap O$ with $O \subseteq \mathbb{R}^N$ open) onto an open set $V \subseteq \mathbb{R}^k$, continuous in both directions.

**Proposition 11.8** (No chart covers the sphere). *Let $S^2_R = \{\mathbf{L} \in \mathbb{R}^3 : |\mathbf{L}| = R\}$ with $R > 0$. There is no chart $\kappa : S^2_R \to V$ with $V \subseteq \mathbb{R}^2$ open.*

*Proof.* The first component $\kappa_1$ of $\kappa$ is a continuous function on the closed, bounded set $S^2_R \subseteq \mathbb{R}^3$, so it attains a maximum at some point $P$. But $V$ is open, so it contains $\kappa(P) + (\varepsilon, 0)$ for small $\varepsilon > 0$, and this is $\kappa$ of a point where $\kappa_1$ is larger. ∎

As in Lecture 2, we use overlapping coordinate patches. Here their transition maps must be *canonical*.

**Definition 11.9** (Phase space). Let $\mathcal{M} \subseteq \mathbb{R}^N$. A *canonical atlas* of dimension $2n$ on $\mathcal{M}$ is a family of charts $\kappa_a : U_a \to V_a \subseteq \mathbb{R}^{2n}$ whose domains cover $\mathcal{M}$, such that for every pair $a, b$ the *transition map* $\tau_{ba} = \kappa_b \circ \kappa_a^{-1} : \kappa_a(U_a \cap U_b) \to \kappa_b(U_a \cap U_b)$ is smooth and canonical (Definition 8.1). The coordinates $(q, p)$ of a chart are *canonical coordinates*. A set with a canonical atlas is a *phase space*, or *symplectic manifold*.

Now the sphere. Away from the poles $(0,0,\pm R)$ use Archimedes' coordinates, the *azimuth* $\varphi$ and the *height* $z = L_3$:

$$\mathbf{L} = \big(\rho\cos\varphi,\ \rho\sin\varphi,\ z\big), \qquad \rho = \sqrt{R^2 - z^2}, \qquad |z| < R. \tag{11.7}$$

An angle needs two ranges, $(-\pi,\pi)$ and $(0,2\pi)$, to reach every azimuth, so "the chart $(\varphi, z)$" is really two charts differing by the shift $\varphi \mapsto \varphi + 2\pi$. A second pair of charts uses the first axis: the height is $L_1$, and $\psi$ is the angle about the first axis measured from $e_2$ towards $e_3$, so that $\mathbf{L} = (L_1, \rho\cos\psi, \rho\sin\psi)$ with $\rho = \sqrt{R^2 - L_1^2}$. The resulting chart is the first one composed with the cyclic relabelling $(L_1, L_2, L_3) \mapsto (L_2, L_3, L_1)$, and it misses only $(\pm R, 0, 0)$, so between them these charts cover the sphere.

**Theorem 11.10** (Canonical coordinates on the Lie–Poisson spheres · *proved; the completeness in (1) uses T1 (cited)*)**.** *Let $R > 0$. For the Lie–Poisson bracket (11.4):*

1. *every Hamiltonian flow maps $S^2_R$ to itself and is complete there;*
2. *for $\mathbf{L} \in S^2_R$ and $f, g$ smooth near $\mathbf{L}$, the value $\{f,g\}(\mathbf{L})$ depends only on the restrictions of $f$ and $g$ to the sphere through $\mathbf{L}$;*
3. *in the chart $(\varphi, z)$ this bracket is the canonical bracket of $q = \varphi,\ p = z$, so that*

$$\{\varphi, z\} = +1, \tag{11.8}$$

   *and in the chart $(\psi, L_1)$ it is the canonical bracket of $q = \psi,\ p = L_1$. These charts are therefore a canonical atlas, and $S^2_R$ is a phase space of dimension 2;*
4. *$\Pi(0) = 0$: every $X_f$ vanishes at the origin, which is therefore an orbit by itself.*

*So $\mathbb{R}^3$, which is odd-dimensional and has no canonical coordinates, is the disjoint union of the spheres and the origin, and on each piece the Lie–Poisson bracket is the bracket of a phase space. These pieces are the* symplectic leaves*, and here the Casimir $|\mathbf{L}|^2$ labels them.*

*Proof.* (1) $|\mathbf{L}|^2$ is a Casimir (Proposition 11.7), so it is conserved by every flow, and a motion that starts on $S^2_R$ stays on it as long as it exists; $S^2_R$ is compact, so by the blow-up alternative of Theorem 1.3 the motion exists for all time, and Theorem 1.5 makes the flow a one-parameter group of maps of $S^2_R$.

(2) $\mathbf{L}\cdot(\nabla f \times \nabla g)$ is unchanged if $\nabla f$ or $\nabla g$ is altered by a multiple of $\mathbf{L}$. Two smooth functions that agree on the sphere through $\mathbf{L}$ have gradients at $\mathbf{L}$ that differ by such a multiple, since their difference vanishes on the sphere.

(3) The azimuth $\varphi$ and the height $L_3$ are also smooth functions on an open subset of $\mathbb{R}^3$, with $\nabla\varphi = (-L_2, L_1, 0)/(L_1^2 + L_2^2)$, so by (11.4)

$$\{\varphi, L_3\} = \mathbf{L}\cdot\big(\nabla\varphi \times e_3\big) = \mathbf{L}\cdot\frac{(L_1, L_2, 0)}{L_1^2 + L_2^2} = 1.$$

Let $f, g$ have local expressions $F(\varphi, z)$, $G(\varphi, z)$ on the sphere. By (2) we may replace $f$ by $F(\varphi, L_3)$ and $g$ by $G(\varphi, L_3)$, and then the chain rule and antisymmetry give

$$\{f,g\} = \left(\frac{\partial F}{\partial\varphi}\frac{\partial G}{\partial z} - \frac{\partial F}{\partial z}\frac{\partial G}{\partial\varphi}\right)\{\varphi, L_3\} = \frac{\partial F}{\partial\varphi}\frac{\partial G}{\partial z} - \frac{\partial F}{\partial z}\frac{\partial G}{\partial\varphi},$$

the canonical bracket of $F$ and $G$. The same computation with the indices relabelled cyclically, $1 \to 2 \to 3 \to 1$, which leaves $\varepsilon_{abc}$ unchanged, gives $\{\psi, L_1\} = 1$ and the same conclusion in the second chart.

For the atlas, the two ranges of $\varphi$ (or of $\psi$) differ by a translation, which is canonical. The transition $\tau$ from $(\varphi, z)$ to $(\psi, L_1)$ is smooth, and its two components are the local expressions of $\psi$ and $L_1$ in the first chart. By what was just shown their canonical bracket is $\{\psi, L_1\} = 1$. This is the coordinate-bracket condition of Theorem 8.2 for $n = 1$, so $\tau$ is canonical. The transition the other way is canonical by the same argument with $\{\varphi, L_3\} = 1$.

(4) is (11.4) at $\mathbf{L} = 0$. ∎

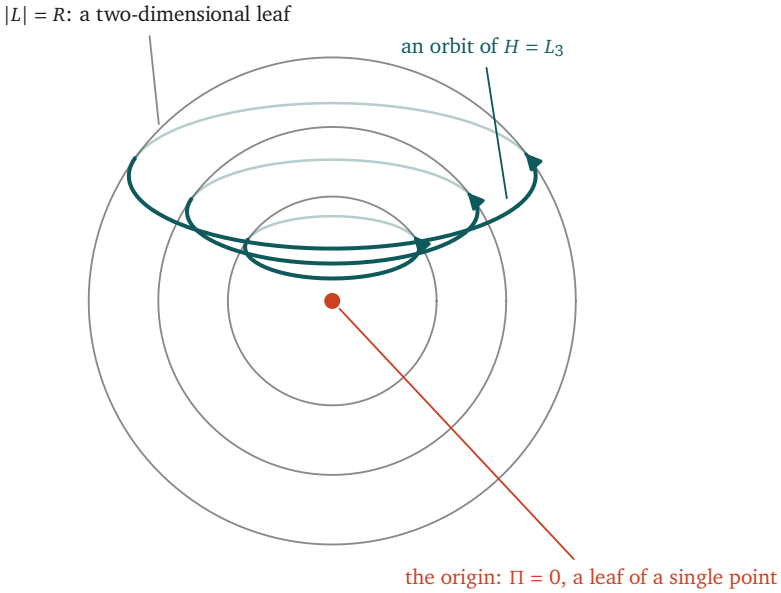



Figure 11.1: The symplectic leaves of (11.4): the spheres $|\mathbf{L}| = R$, each a two-dimensional phase space, and the origin, a leaf consisting of one point, where $\Pi = 0$. On each sphere one orbit of $H = L_3$ is drawn; it was integrated from $\dot{\mathbf{L}} = \nabla H \times \mathbf{L}$, not traced as a circle of latitude, and $|\mathbf{L}|$ and $L_3$ are constant along it to $10^{-10}$.

## 11.4 The rigid body and its sign

A rigid body with a point fixed at the origin has configuration space SO(3) (§ 2.2), so its phase space $T^*\mathrm{SO}(3)$ is six-dimensional. Its kinetic energy is unchanged if the whole body is rotated about the fixed point, and for a free body the three components of its spatial angular momentum are conserved (§ 11.5 sketches why). Keeping only the angular momentum leaves three variables and a degenerate bracket on $\mathbb{R}^3$, as $q \times p$ did. We state the result and do not prove it: nothing in this course builds $T^*\mathrm{SO}(3)$.

**Proposition 11.11** (Reduction of the rigid body · *cited*)**.** *Dividing the rotations out of the rigid body's phase space leaves the* body *angular momentum* $\mathbf{M}$ *as its state, with the Poisson structure* $(\mathbb{R}^3, -\Pi)$*, that is,*

$$\{f, g\}(\mathbf{M}) = -\mathbf{M} \cdot (\nabla f \times \nabla g). \tag{11.9}$$

*Read in the* spatial *angular momentum* $\mathbf{L}$ *instead, the bracket is* (11.4)*, as for* $q \times p$*. (Marsden–Ratiu §§13.1–13.4 and §15.3, pp. 489–491; their conventions are ours.)*

*The body bracket carries a minus sign.* The map $\mathbf{L} \mapsto -\mathbf{L}$ is a Poisson map from (11.4) to (11.9). We check the sign independently in § 11.5 by deriving the motion of $\mathbf{M}$ from rigid-body kinematics. Only the body bracket reproduces that equation.

**Corollary 11.12** (The body leaves)**.** *For the body bracket* (11.9)*,* $X_f = \mathbf{M} \times \nabla f$*, and the Casimirs, the leaves and their canonical charts are those of Proposition 11.7 and Theorem 11.10, carried over by* $\mathbf{L} \mapsto -\mathbf{L}$*, which maps each sphere to itself. In particular* $\{\varphi, M_3\} = -1$*: the canonical chart is* $(\varphi, -M_3)$*.*

*Proof.* $X_f = -\Pi \nabla f = -\nabla f \times \mathbf{M}$ by (11.6). A Poisson bijection carries Casimirs to Casimirs, flows to flows and canonical charts to canonical charts, and $-\mathbf{L}$ lies on the sphere of $\mathbf{L}$; and $\varphi(-\mathbf{L}) = \varphi(\mathbf{L}) \pm \pi$, $L_3(-\mathbf{L}) = -L_3(\mathbf{L})$. ∎

## 11.5 Euler's equations

### The body variables

Let the body have a point fixed at the origin and be made of mass points $m_a$ at positions $X_a$ *in the body*, not all on one line through the origin. Its *attitude* at time $t$ is the rotation $R(t) \in \mathrm{SO}(3)$ that carries the reference placement to the current one, so the spatial positions are $x_a(t) = R(t) X_a$. Differentiating $R^{\mathsf{T}} R = \mathbb{1}$ gives $\dot R^{\mathsf{T}} R + R^{\mathsf{T}} \dot R = 0$, so $R^{\mathsf{T}} \dot R$ is antisymmetric, and by Proposition 10.2 there is a unique $\Omega(t) \in \mathbb{R}^3$ — the *body angular velocity* — with

$$R^{\mathsf{T}} \dot R = \hat\Omega, \qquad \hat\Omega u = \Omega \times u, \qquad \text{hence} \qquad \dot x_a = R\,(\Omega \times X_a). \tag{11.10}$$

**Lemma 11.13** (The inertia tensor)**.** $\mathbb{I}\Omega := \sum_a m_a\, X_a \times (\Omega \times X_a)$ *is a symmetric linear map of* $\mathbb{R}^3$*, positive definite because the* $X_a$ *do not all lie on one line through the origin, and*

$$T = \frac{1}{2} \sum_a m_a\, |\dot x_a|^2 = \frac{1}{2}\, \Omega \cdot \mathbb{I}\Omega, \qquad \mathbf{L} := \sum_a m_a\, x_a \times \dot x_a = R\, \mathbb{I}\Omega.$$

*Proof.* By the triple-product identity $u \cdot \mathbb{I} v = \sum_a m_a (u \times X_a) \cdot (v \times X_a)$: symmetric, and positive at $u = v \neq 0$ unless every $X_a$ lies on $\mathbb{R}u$. Then $|\dot x_a| = |\Omega \times X_a|$ gives $T$, and $Ru \times Rv = R(u \times v)$ gives $\mathbf{L}$. ∎

Write $\mathbf{M} := R^{\mathsf{T}} \mathbf{L} = \mathbb{I}\Omega$ for the *body* angular momentum. In principal axes — the orthonormal eigenbasis of $\mathbb{I}$ — $\mathbb{I} = \mathrm{diag}(I_1, I_2, I_3)$ with $I_i > 0$, so

$$M_i = I_i \Omega_i, \qquad T = \frac{1}{2} \sum_i I_i \Omega_i^2 = \frac{1}{2} \sum_i \frac{M_i^2}{I_i}. \tag{11.11}$$

**Proposition 11.14** (Euler's equation, from kinematics)**.** *Let* $R(t) \in \mathrm{SO}(3)$ *be a smooth motion whose spatial angular momentum* $\mathbf{L} = R\,\mathbb{I}\Omega$ *is constant. Then* $\mathbf{M} = R^{\mathsf{T}} \mathbf{L}$ *satisfies*

$$\dot{\mathbf{M}} = \mathbf{M} \times \Omega, \qquad \Omega_i = M_i / I_i. \tag{11.12}$$

*Proof.* $\dot{\mathbf{M}} = \dot R^{\mathsf{T}} \mathbf{L} = (R\hat\Omega)^{\mathsf{T}} \mathbf{L} = \hat\Omega^{\mathsf{T}} R^{\mathsf{T}} \mathbf{L} = -\hat\Omega \mathbf{M} = -\Omega \times \mathbf{M} = \mathbf{M} \times \Omega$, using (11.10) and the antisymmetry of $\hat\Omega$. ∎

The hypothesis is the only physical input, and we sketch why it holds for a free body. Let no external force act except the one that holds the fixed point, which has no torque about that point. The internal forces of a rigid body cancel in pairs along the lines joining the points — Newton's third law in its strong form — so the total torque about the fixed point is zero and $\mathbf{L}$ is constant.

### Euler's equations are Lie–Poisson

**Theorem 11.15** (Euler's equations). *On $\mathbb{R}^3$ with the body bracket (11.9) and*

$$H(\mathbf{M}) = \frac{1}{2}\Big(\frac{M_1^2}{I_1} + \frac{M_2^2}{I_2} + \frac{M_3^2}{I_3}\Big),$$

*the evolution $\dot f = \{f, H\}$ is exactly (11.12), that is*

$$\dot M_1 = \frac{I_2 - I_3}{I_2 I_3} M_2 M_3, \qquad \dot M_2 = \frac{I_3 - I_1}{I_3 I_1} M_3 M_1, \qquad \dot M_3 = \frac{I_1 - I_2}{I_1 I_2} M_1 M_2. \tag{11.13}$$

*The spatial bracket (11.4) and the same $H$ would give $\dot{\mathbf{M}} = \Omega \times \mathbf{M}$, the time reverse.*

*Proof.* By Corollary 11.12 the Hamiltonian field is $\mathbf{M} \times \nabla H$, and $\nabla H = \Omega$ by (11.11). Writing out $\mathbf{M} \times \Omega$ componentwise gives (11.13). With $\Pi$ in place of $-\Pi$ the field is $\nabla H \times \mathbf{M} = \Omega \times \mathbf{M}$ by (11.6). ∎

So the free rigid body is a Hamiltonian system on $\mathbb{R}^3$, with a degenerate bracket and three equations in place of six. It has two conserved quantities of different kinds: the Casimir $|\mathbf{M}|^2$, conserved because of the bracket, and $H$, conserved because it is the Hamiltonian. The motion runs along the intersection of a sphere with an energy ellipsoid.

**Proposition 11.16** (The six equilibria). *Let $I_1 < I_2 < I_3$ and $R > 0$. The equilibria of (11.12) on the leaf $|\mathbf{M}| = R$ are exactly the six points $\pm Re_1, \pm Re_2, \pm Re_3$: the rotations about the principal axes.*

*Proof.* The $a$-th component of $\mathbf{M} \times \Omega$ is $\big(1/I_c - 1/I_b\big) M_b M_c$ with $(a, b, c)$ cyclic. The three moments are distinct, so all three components vanish if and only if $M_b M_c = 0$ for each of the three pairs, that is, if and only if at most one $M_i$ is non-zero. On $|\mathbf{M}| = R$ that leaves the six listed points. ∎

## 11.6 The stable axes

Each leaf is a two-dimensional phase space, so on a leaf the rigid body has one degree of freedom, and the question of Lecture 1 can be asked: which equilibria are stable? Throughout this section and the next $I_1 < I_2 < I_3$ and $R > 0$, and "stable on the leaf" is Definition 1.13 with neighbourhoods taken in the sphere $S_R^2$.

**Theorem 11.17** (The axes of least and greatest inertia are stable on the leaf · *proved; the completeness uses T1 (cited)*). *On $S_R^2$ the function $H$ attains its maximum $R^2/(2I_1)$ exactly at $\pm Re_1$ and its minimum $R^2/(2I_3)$ exactly at $\pm Re_3$. Each of these four points is a stable equilibrium on the leaf.*

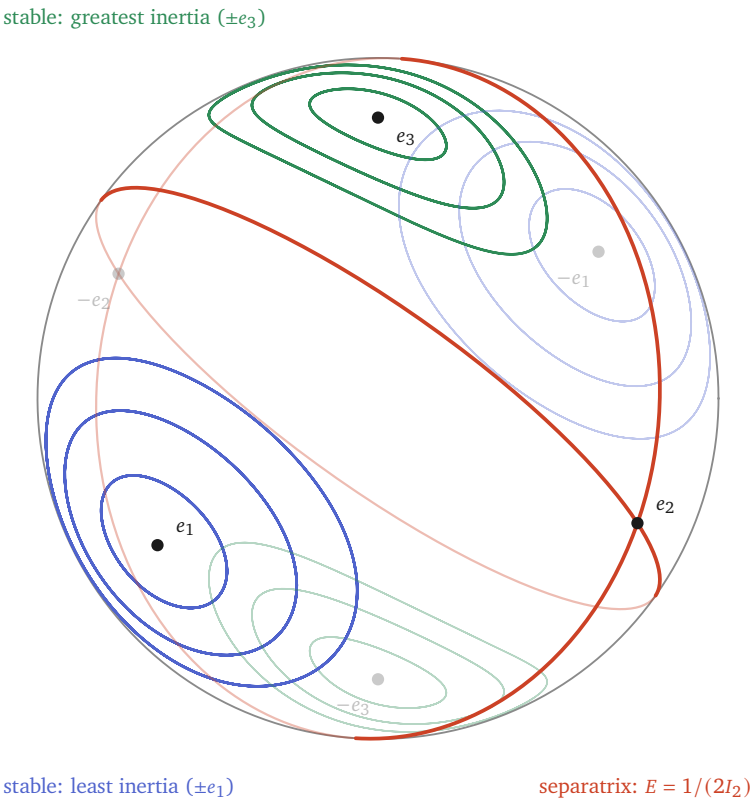



Figure 11.2: The free rigid body on the leaf $|\mathbf{M}| = 1$, with $I = (1, 2, 3)$. Every blue and green curve is an orbit, obtained by integrating (11.12) and not by intersecting the two quadrics; $|\mathbf{M}|$ and $H$ are constant along each to $10^{-8}$. Blue: the orbits round $\pm e_1$, the axis of *least* inertia, where $H$ is largest. Green: the orbits round $\pm e_3$, of *greatest* inertia, where $H$ is smallest. Red: the level $H = 1/(2I_2)$, two great circles drawn from their own equation $M_3 = \pm kM_1$ (§ 11.7), both passing through the two equilibria $\pm e_2$. The far hemisphere is faint.

*Proof.* On $|\mathbf{M}| = R$,

$$2H - \frac{R^2}{I_1} = \sum_i M_i^2\Big(\frac{1}{I_i} - \frac{1}{I_1}\Big) = M_2^2\Big(\frac{1}{I_2} - \frac{1}{I_1}\Big) + M_3^2\Big(\frac{1}{I_3} - \frac{1}{I_1}\Big) \le 0,$$

with equality only if $M_2 = M_3 = 0$, that is at $\pm Re_1$; and symmetrically $2H - R^2/I_3 = M_1^2(1/I_1 - 1/I_3) + M_2^2(1/I_2 - 1/I_3) \ge 0$ with equality only at $\pm Re_3$. The four points are equilibria by Proposition 11.16.

Stability is Dirichlet's argument: a conserved function with a strict extremum traps the motion. Take $x_* = Re_1$ and $\varepsilon \in (0, 2R)$; a smaller $\delta$ serves a larger $\varepsilon$, so this is no restriction. The set $K = \{\mathbf{M} \in S_R^2 : |\mathbf{M} - x_*| = \varepsilon\}$ is closed and bounded and contains neither $Re_1$ nor $-Re_1$, so $H < R^2/(2I_1)$ on $K$ and its maximum $m$ there is still less than $R^2/(2I_1)$. By continuity choose $\delta \in (0, \varepsilon)$ with $H > m$ at every point of the sphere within $\delta$ of $x_*$. A motion starting there remains on the sphere for all time by Theorem 11.10 and Corollary 11.12. Since $H$ is conserved, $H > m$ for all $t$ and the motion never meets $K$. Its distance from $x_*$ is continuous in $t$ and starts below $\varepsilon$, so it stays below $\varepsilon$ for all $t \ge 0$. The same argument applies at $-Re_1$ and, with reversed inequalities, at $\pm Re_3$. ∎

## 11.7 The unstable axis

At the middle axis $H$ has neither a maximum nor a minimum on the sphere. On $|\mathbf{M}| = R$,

$$2H - \frac{R^2}{I_2} = M_1^2\Big(\frac{1}{I_1} - \frac{1}{I_2}\Big) - M_3^2\Big(\frac{1}{I_2} - \frac{1}{I_3}\Big),$$

which takes both signs arbitrarily near $\pm Re_2$ and vanishes exactly on the two great circles $\Gamma_\pm = \{|\mathbf{M}| = R,\ M_3 = \pm k M_1\}$, with $k = \sqrt{(1/I_1 - 1/I_2)/(1/I_2 - 1/I_3)} > 0$, both through $\pm Re_2$ (Figure 11.2).

**Corollary 11.18** (The middle axis is unstable · *sketched*)**.** *Neither $+Re_2$ nor $-Re_2$ is a stable equilibrium on the leaf.*

*Sketch.* On $\Gamma_-$, (11.13) gives $\dot{M}_2 = (1/I_1 - 1/I_3) M_3 M_1 = -(1/I_1 - 1/I_3) k\, M_1^2$, which is negative except where $M_1 = 0$, that is, at $\pm Re_2$. So a motion that starts on $\Gamma_-$ near $Re_2$ — where it stays, since it keeps the value of $H$ and can pass to $\Gamma_+$ only through an equilibrium — moves away from $Re_2$ with $M_2$ decreasing. That it then tends to $-Re_2$ is the step not proved here ($M_2$ has a limit, which can only be at an equilibrium); granting it, starting points arbitrarily close to $Re_2$ lead arbitrarily close to $-Re_2$, at distance $2R$. On $\Gamma_+$ the same holds with the signs reversed, at $-Re_2$. ∎

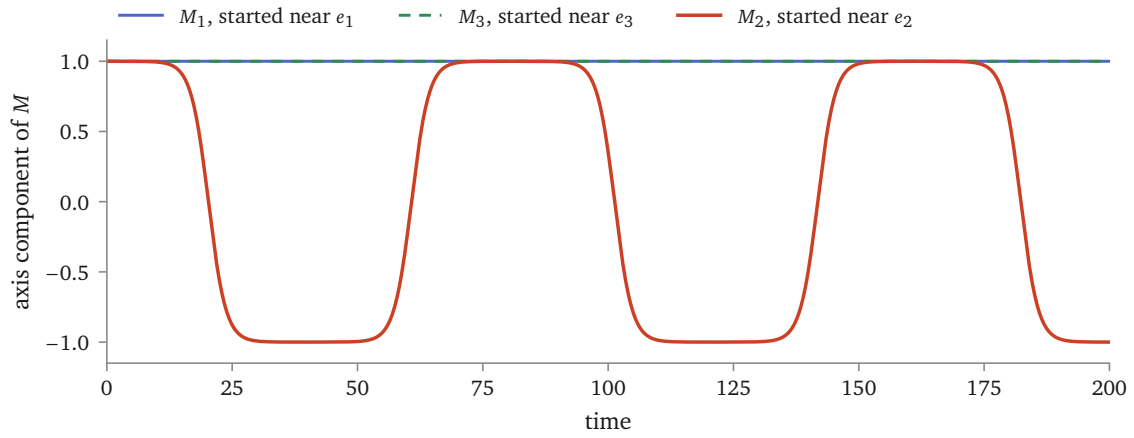



Figure 11.3: $I = (1, 2, 3)$, $|\mathbf{M}| = 1$, each run started close to a principal axis. The axis component of $\mathbf{M}$ stays close to 1 for the two extreme axes (the $e_3$ run dashed, since the two curves nearly coincide); about the middle axis it changes sign five times in 200 time units.

Near $Re_2$ but off $\Gamma_\pm$ the motion runs close beside them, from near one middle-axis equilibrium to near the other and back (Figure 11.3). In space $\mathbf{L}$ stays fixed and the body turns over, repeatedly. This is the *tennis-racket* effect. A racket or a book thrown spinning nearly about its middle axis turns over in the air if the flight lasts long enough; about the other two axes it does not.

**Summary.**

- *Handed on:* Poisson structures (Definition 11.1); the Lie–Poisson bracket and its Casimirs (Proposition 11.7); its leaves, each sphere a phase space covered by canonical charts and by no single chart (Definition 11.9, Theorem 11.10, and Proposition 11.8); the free rigid

body, $\dot{\mathbf{M}} = \mathbf{M} \times \Omega$ for the body bracket (Theorem 11.15), on each sphere stable about its extreme axes and not about the middle one (Theorem 11.17 and Corollary 11.18).

- *Next:* Lecture 12 asks when enough conserved quantities make a motion integrable, and gives the last answer to the course's question.

## References

Marsden and Ratiu, *Introduction to Mechanics and Symmetry*, whose conventions are ours: §1.2 (pp. 7–9) for the rigid-body bracket and Euler's equations; Chapter 10 (p. 327 on) for Poisson manifolds, with Casimir functions on p. 335; §15.3 (pp. 489–491) for the free rigid body and the flow on the angular-momentum spheres (Figure 15.3.1). Arnold, *Mathematical Methods of Classical Mechanics*, Appendix 14, p. 456: here "the leaves are concentric spheres and their center at the origin". Tong, *Lectures on Classical Dynamics*, §3.4.3 (pp. 57–58), for the tennis racket.